



Squid-Inspired Robot

British scientists unveiled a squid-inspired robot that could safely explore and restore endangered coral reefs. <http://digit.in/feb21-07>



Predicting Autism

A series of human sperm biomarkers can predict chances of autism in offsprings with up to 90 percent efficacy. <http://digit.in/feb21-08>

First laser probes are used to illuminate a finger. When the beam touches the veins it makes a “poof” sound which is then picked up by an ultrasound detector. The combined data is used to reconstruct a three dimensional map of the veins in the person’s hand. The entire setup integrates the acoustic and light windows, and has an imaging window similar to that of conventional hand scanners. The system is expected to be used in critical facilities such as bans and military bases.

Researchers from the University of Missouri have suggested reading the veins in the eyes of a person as a biometric mode that can provide additional security, and improve the performance to existing iris scanning systems. As in the hands, the veins in the eyes also have unique patterns.

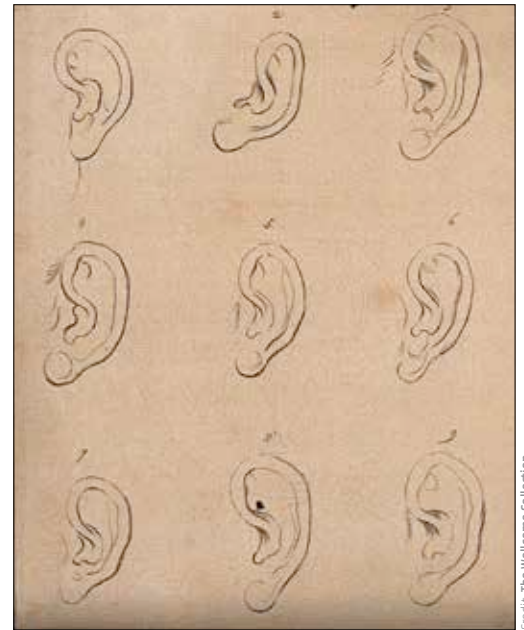
EAR BIOMETRICS

The specific morphologies of the ear are as unique to each individual as fingerprints, and in fact there are police databases around the world that keep records of ear impressions of violent criminals. The approach even predates the use of fingerprints to identify people by about a century, and was actually used way back in Johann Kaspar Lavater, an eighteenth century Swiss poet and physiognomist. Lavater began the field of otomorphology by using ear patterns to distinguish between people. He even went a step ahead and claimed that by precise measurements of a person’s eyes, ears, nose, chin and shape of the face, he could even

predict criminal behavior! This old and obscure idea of using ears as a biometric indicator has found new applications in the era of touchscreen smartphones. This is because the ears are a part of the natural motion of using the phone, and the action of taking a call can be used to unlock the phone. Ergo by Descartes Biometrics is a security solution that does just this. The interface for the biometric identification is the touchscreen itself, as each ear has unique touchpoints. The app can be made more sensitive or less sensitive, which is a tradeoff on convenience. There is no additional hardware required, and the profile for the recognition is stored locally on the device itself.

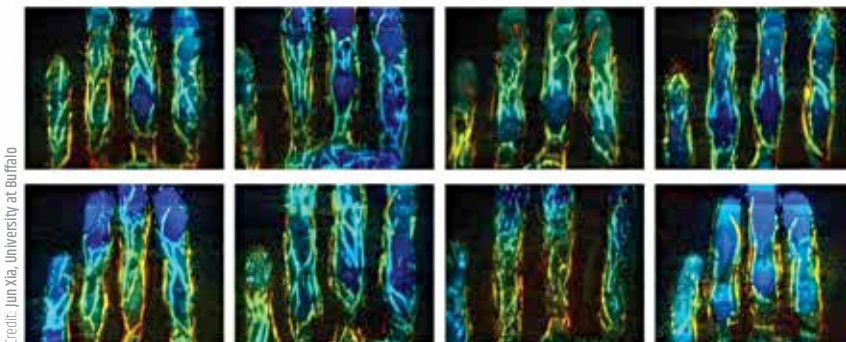
BLINK PATTERNS

Blink Fish originally started as a game where you blink to make a fish jump at just the right timings, threw in a video recording mode and became a rage on TikTok. The app works because smartphones already have the capability to capture and track at least some of the wide range of movements and actions used by your eyes, and use them as inputs. Going forward, we can expect more smartphones and X reality setups to incorporate eye actions and gestures into the ways users interact with these systems. It is not surprising that



A drawing of nine ears by Johann Kaspar Lavater

researchers have attempted to use blink patterns as a biometric indicator. Note here that although the mode of access is biometric, in practice has some similarities to entering a PIN code, and is not entirely dependent on a passive reading, but there are characteristics observed that are uniquely personal in the process. Essentially, the individual who wishes to access the device has to first trigger the playback of the song, and then... to put it in nontechnical terms, blink to the beat. The individual is free to choose their own blink patterns. The software system can then match the blinks on a new unlock attempt, to the pattern stored on the file. This is a strange authentication mechanism as it is immune to shoulder surfing, but someone looking straight at your face and not at the screen can catch the pattern. However, there is a second layer of authentication, one that compares the individual characteristics of the blink itself, to what is known as a personal “blinkprint” saved on file. The technology has been developed by researchers from Georgia Tech. This is one of the contact free biometric authentication methods that may be particularly useful in a post COVID world. 



Finger vessels from eight different subjects shown here

Credit: Jun Xia, University at Buffalo

Credit: The Wellcome Collection